TEXORA · COURSE NOTES · LECTURE SHEET

Basic Ideas of Spinning, Weaving, Knitting and Garments Manufacturing Processes

The four manufacturing families of the textile chain — what each one does, the sequence of operations inside it, and how the four differ from each other.

TE-1109 · FUNDAMENTALS OF TEXTILE ENGINEERING · TOPIC 2 · LEVEL 1 TERM 1

1. Introduction: four ways of building a textile product

Topic 1 ended with the textile chain: fibre → spinning → fabric making → wet processing → garment making → finished product. This topic looks inside the manufacturing stages of that chain and takes the basic idea of each one. There are four manufacturing families:

- **Spinning** converts loose fibres into yarn.
- **Weaving** converts yarns into fabric by interlacing two sets of yarns at right angles.
- **Knitting** converts yarn into fabric by forming and interlocking loops.
- **Garments manufacturing** converts finished fabric into a wearable or usable product by cutting and sewing.

Each family is a full technology in its own right, and each has dedicated courses later in this degree. What follows is the map of each territory: the purpose of the process, the order of its main operations, and the terms without which the later courses cannot be followed.

2. Spinning: fibre into yarn

2.1 What spinning is

Spinning is the process of converting a mass of short, disordered fibres into a continuous, uniform strand of yarn by opening, cleaning, aligning, drafting (drawing out) and twisting the fibres. Loose fibres have almost no usable strength as a mass; a yarn is strong because its fibres are laid parallel and twisted so that they press against and grip one another. Friction between the fibres, created by twist, is what holds the yarn together.

2.2 The objects of spinning

- **Opening** — breaking up the compressed bales of fibre into small tufts so that dirt can be removed and fibres can be mixed evenly.
- **Cleaning** — removing trash, dust, seed fragments and other impurities from the fibres.
- **Blending** — mixing fibres of different lots (or different types, such as cotton with polyester) so that the yarn is uniform and has the required properties at the planned cost.
- **Parallelisation** — straightening the fibres and laying them parallel to one another, removing hooks and curls.
- **Drafting** — drawing the strand out, stage by stage, until it becomes fine enough to be the yarn.
- **Twisting** — inserting twist to give the strand strength and cohesion.
- **Winding** — winding the finished yarn onto a package (cone or cheese) suitable for the next process.

2.3 The sequence of machines (cotton ring spinning)

The classic sequence for spinning cotton into ring-spun yarn is:

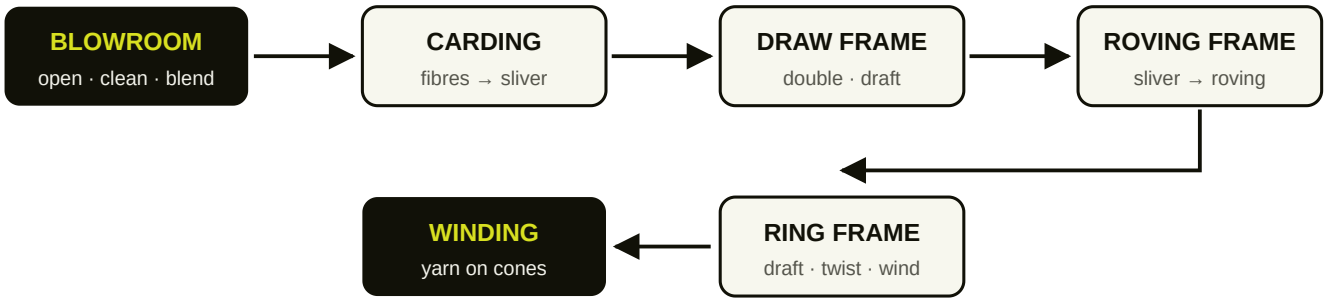


Figure 1 — The machine sequence of cotton ring spinning, from bale to yarn package.

- **Blowroom.** A line of opening and cleaning machines that turns bale fibre into a uniform sheet (lap) or feeds opened fibre directly to the cards.
- **Carding.** The heart of spinning: wire-covered surfaces individualise the fibres, remove most of the remaining trash and short fibres, and deliver the fibres as a soft rope called **sliver**.
- **Draw frame.** Several slivers (usually six or eight) are combined (*doubling*) and drafted together, which averages out thick and thin places and straightens the fibres.
- **Roving frame (simplex).** The sliver is drafted much finer and given a slight twist to form **roving**, wound on bobbins small enough for the final spinning frame.
- **Ring frame.** The roving is drafted to the final fineness, twisted to the required strength and wound onto small bobbins (cops). This is where the yarn is truly “made”.
- **Winding.** The small cops are rewound into large cones or cheeses, and yarn faults are cut out and rejoined (clearing and splicing) on the way.

For medium and coarse yarns a shorter route is common: **rotor (open-end) spinning** feeds sliver directly into a high-speed rotor that drafts and twists it in one step, skipping the roving and ring frames. Rotor yarn is a little weaker and harsher but much cheaper to produce. Fine and high-quality yarns still come from the ring frame, often with an added **comber** stage before the draw frame, which removes short fibres and produces the smoothest, strongest yarns.

> **KEY POINT**
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> Spinning is a process of gradual refinement. No single machine turns fibre into yarn; instead, each stage makes the strand finer, cleaner, more parallel and more even than the stage before, and the final twist locks the structure into yarn.

3. Weaving: yarn into fabric by interlacing

3.1 What weaving is

Weaving is the process of interlacing two sets of yarns at right angles to each other on a machine called a loom. The lengthwise set, held under tension on the loom, is the **warp**; the crosswise yarn inserted through it, one row at a time, is the **weft** (also called the pick). Because the two sets are interlaced rather than merely laid together, the resulting fabric is firm, stable and strong — the classic structure of shirting, denim, towels and suiting.

3.2 Weaving preparation

Weft yarn can go to the loom almost as it comes from spinning, but warp yarn must survive thousands of tension and abrasion cycles, so it is prepared carefully:

- **Winding** — warp yarn is rewound onto large, fault-free packages.
- **Warping** — hundreds of yarn ends are arranged side by side in a sheet and wound onto a warper’s beam, all at equal tension.
- **Sizing** — the warp sheet is coated with a protective adhesive film (traditionally starch) that glues down protruding fibres and strengthens the yarn against loom abrasion; the size is removed later in wet processing (desizing).
- **Drawing-in / tying-in** — each warp end is passed through the loom’s heald eyes and reed dents in the order the weave pattern requires.

3.3 The loom motions

Whatever its size or speed, every loom performs the same five primary and secondary motions, in the same repeating order — one cycle produces one weft pick:

MOTION	WHAT IT DOES
Shedding	Warp ends are separated into two layers, opening a tunnel (the shed) through which the weft will pass. Which ends are lifted decides the weave pattern.
Picking	One weft yarn (pick) is inserted across the shed — by a shuttle, projectile, rapier, air jet or water jet.
Beat-up	The reed swings forward and pushes the newly inserted pick into the edge of the cloth already formed (the cloth fell).
Let-off	Warp yarn is released from the warp beam at a controlled rate to feed the process.
Take-up	The woven cloth is wound onto the cloth roller at a rate that sets the weft density (picks per unit length).

Modern shuttleless looms insert weft at very high rates, so a loom can produce many metres of fabric per hour; the motions, however, are exactly the same as on a handloom. What machinery changed was speed, width and consistency — not the principle.

4. Knitting: yarn into fabric by looping

4.1 What knitting is

Knitting is the process of forming fabric by creating loops of yarn and drawing new loops through previously formed loops, so that the loops interlock and hold one another. The fabric is built loop by loop, most commonly by rows of needles working together. Because the structure is loops rather than a grid, knitted fabric stretches, recovers, drapes softly and feels comfortable next to the skin — the structure of T-shirts, sweaters, socks and underwear.

4.2 The two families of knitting

FEATURE	WEFT KNITTING	WARP KNITTING
Yarn path	One yarn (or a few) travels across the fabric, forming loops course by course	Many yarns travel along the fabric, each forming loops in the lengthwise direction
Typical machines	Circular knitting machines, flat-bed machines	Tricot and raschel machines
Typical products	T-shirts, sweaters, socks, fleece	Lace, nets, sportswear, upholstery
Behaviour	Very elastic; can unravel (ladder) from the last course	More stable; does not ladder or unravel easily

4.3 The basic elements

On a knitting machine, three elements cooperate to form every loop. The **needle** (usually a latch needle) catches the new yarn and pulls it through the old loop. The **sinker** holds the fabric down and helps form the loop. The **cam** system drives the needles up and down in the correct order as the machine runs. Machine fineness is expressed as **gauge** — the number of needles per unit length — which, together with yarn count, fixes how fine the fabric can be. In circular weft knitting, dozens of yarn feeders work around a rotating cylinder of needles, and fabric emerges continuously as a tube.

> **WORKED COMPARISON — THE SAME YARN, TWO FABRICS**
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> Knitted single jersey and woven plain-weave fabric can be made from the very same cotton yarn, yet they behave quite differently. The jersey stretches easily in both directions, curls at the edges and feels soft — ideal for a T-shirt. The woven fabric keeps its dimensions, resists stretching and has a crisper surface — ideal for a formal shirt. Structure, not fibre, creates most of the difference: interlacing locks yarns in place, while loops let them move.

5. Garments manufacturing: fabric into product

5.1 What the process includes

Garments manufacturing (apparel manufacturing) converts finished fabric into a complete garment through the operations of design and pattern making, cutting, sewing and finishing. Unlike spinning, weaving and knitting, it works garment by garment rather than by the metre, and it remains the most labour-intensive stage of the whole chain.

5.2 The sequence of operations

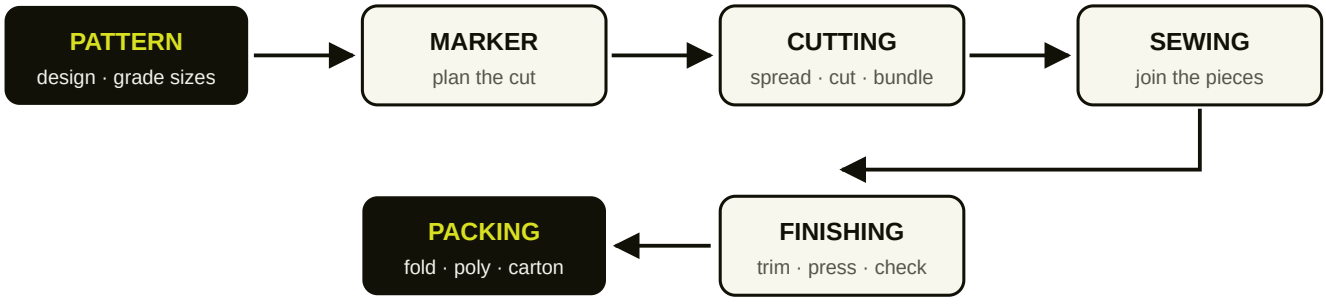


Figure 2 — The main operations of garments manufacturing.

- **Pattern making and grading.** A pattern — the template of every piece of the garment — is drafted for a base size, then *graded* up and down to produce the full size range.

- **Marker making.** All the pattern pieces for an order are arranged on a plan (the marker) whose length and width match the fabric, aiming to waste as little fabric as possible; marker efficiency is watched closely because fabric is the largest cost in a garment.
- **Spreading and cutting.** Fabric is spread in many layers (a lay), the marker is placed on top, and the pieces are cut out with straight-knife, band-knife or computer-controlled cutting machines. Cut pieces are numbered and bundled by size and shade.
- **Sewing.** The pieces are joined in a planned order of operations on industrial sewing machines — lockstitch for main seams, overlock for edge neatening, flatlock for knitwear, plus specialised machines for buttonholes, buttons and bartacks.
- **Finishing.** Loose threads are trimmed, the garment is washed if required, pressed, measured and inspected against the buyer's specification.
- **Packing.** Approved garments are folded, tagged, poly-bagged and cartoned for shipment.

Sewing is organised on lines: each operator performs one or a few operations and passes the garment on, so a line of thirty operators and machines can complete hundreds of garments a day. Balancing that line — arranging operations so no operator is idle or overloaded — is one of the central skills of apparel production, studied in the apparel manufacturing courses.

6. The four processes compared

PROCESS	TAKES IN	GIVES OUT	BASIC PRINCIPLE
Spinning	Loose fibres	Yarn	Drafting and twisting fibres into a continuous strand
Weaving	Warp and weft yarns	Woven fabric	Interlacing two yarn sets at right angles
Knitting	Yarn	Knitted fabric	Interlocking loops of yarn
Garments manufacturing	Finished fabric	Garment / product	Cutting fabric pieces and sewing them together

7. Yarn count — the number every process quotes

One measurement ties all four processes together: the fineness of yarn, its **count**. In the direct system (tex), count is the weight per unit length — a 20 tex yarn weighs 20 g per 1,000 m, and a bigger number means a coarser yarn. In the indirect system used for cotton (English count, Ne), count is the number of standard hanks (840 yards) weighing one pound — here a bigger number means a *finer* yarn, the opposite convention, and the source of endless beginner confusion. The two are linked by tex ≈ 590.5 ÷ Ne, so a Ne 30 cotton yarn is about 19.7 tex. Spinning is planned in counts, weaving and

knitting consume yarn by count, and a garment's fabric weight (GSM) is largely decided by the counts chosen — which is why this single number appears in every process of this topic.

8. Glossary — key terms of this topic

TERM	MEANING
Sliver	The soft, thick, untwisted rope of fibres delivered by the card and handled by the draw frame.
Roving	A drafted, slightly twisted strand of fibres, finer than sliver, fed to the ring frame.
Drafting	Drawing a strand out between rollers running at increasing speeds, making it finer and the fibres more parallel.
Doubling	Feeding several strands together so that their irregularities average out.
Sizing	Coating warp yarn with a protective adhesive film so it can withstand weaving.
Shed	The opening formed between raised and lowered warp ends, through which the weft passes.
Pick	One length of weft yarn inserted across the warp in one loom cycle.
Course / wale	In knitting, a horizontal row of loops (course) and a vertical column of loops (wale).
Gauge	The fineness of a knitting machine: the number of needles per unit length.
Marker	The plan showing how all pattern pieces are laid on the fabric for cutting with minimum waste.
Lay	The stack of fabric layers spread for cutting several garments at once.

9. Key points and summary

- Spinning turns fibres into yarn by opening, cleaning, blending, parallelising, drafting and twisting; the cotton sequence is blowroom → carding → draw frame → roving frame → ring frame → winding.
- Weaving interlaces warp and weft at right angles. Warp yarns are wound, warped, sized and drawn in before the loom repeats its five motions — shedding, picking, beat-up, let-off and take-up — once per pick.
- Knitting forms fabric from interlocking loops. Weft knitting (circular and flat machines) is elastic and can ladder; warp knitting (tricot and raschel) is more stable.

- Garments manufacturing turns fabric into products: pattern → grading → marker → spreading → cutting → sewing → finishing → packing.
- Structure explains behaviour: twisted strands give yarn its strength, interlacing gives woven fabric its stability, and loops give knitted fabric its stretch.

10. Practice

Answer in your own words; brief model answers follow.

1. Put the ring-spinning sequence in order and name the material form after each machine: card, ring frame, draw frame, blowroom, roving frame.
2. Weaving needs two yarn systems; knitting usually needs one. Name weaving's two systems and the preparation each receives.
3. A buyer's order is for stretchy T-shirts at low cost in huge quantity. Choose between weaving and knitting for the body fabric and justify the choice in two points.
4. Convert: a cotton yarn of Ne 40 into tex, and a yarn of 25 tex into Ne (use tex ≈ 590.5 ÷ Ne).
5. In the garment sequence of Figure 2, why must marker making come before spreading and cutting, and what is wasted if the marker is inefficient?
6. State one structural difference between rotor-spun and ring-spun yarn that a buyer might feel in a fabric.

Answer key

1. Blowroom (lap/chute feed) → card (sliver) → draw frame (drawn sliver) → roving frame (roving) → ring frame (yarn).
2. Warp — wound on a beam, sized for strength; weft — supplied on pirns/cones, usually unsized.
3. Knitting: loop structure gives the stretch a T-shirt needs, and circular knitting produces fabric fast and cheaply at volume without warp preparation.
4. Ne 40 → 590.5 ÷ 40 ≈ 14.8 tex. 25 tex → Ne ≈ 590.5 ÷ 25 ≈ 23.6, i.e. about Ne 24.
5. The marker fixes how pattern pieces nest on the cloth; spreading and cutting follow its layout. An inefficient marker wastes fabric — the largest single cost in a garment.
6. Rotor yarn is typically bulkier and less strong with a harsher, more matte handle; ring yarn is finer, stronger and smoother (accept equivalent correct contrasts).
